Main Components

- Privileged Peer (Admin)
- Cluster Leader
- Peer Node (Student PC)
- P2P Network Layer
- Security Layer
- Local Storage
- Assignment Metadata Store

Instead of one central server:

- Every lab computer (peer) can receive & share data
- Peers are grouped into clusters
- One peer per cluster temporarily acts as Cluster Leader
- A Privileged Peer (Admin) controls authentication and approval
- Data is transferred in chunks, verified by hashes
- System works even if some peers go offline

Workflow is as follows

1. Peer joins network
2. Authentication & authorization
3. Cluster assignment
4. Cluster leader selection
5. File chunking
6. Peer-to-peer chunk distribution
7. Integrity verification
8. Inter-cluster synchronization
9. Offline storage & later sync

Step 1: Peer Discovery

- Peers discover each other in the LAN
- Privileged Peer is identified
- Available peers broadcast basic status

Step 2: Authentication

- Peer requests permission to join
- Privileged Peer verifies identity
- Approved peers are allowed to participate

Step 3: Adaptive Clustering

- Peers grouped based on:
  - Network proximity
  - Availability
- Each cluster selects a **Cluster Leader**

Step 4: File Preparation (Chunking)

- Large files (software / assignments) are:
  - Split into **small chunks**
  - Each chunk gets a **hash value**
- One or more peers act as **Seed Peers**

Step 5: Chunk-Based Distribution

- Seed Peer sends chunks to cluster leaders
- Cluster leaders distribute chunks inside their cluster
- Peers exchange missing chunks among themselves

Step 6: Integrity Verification

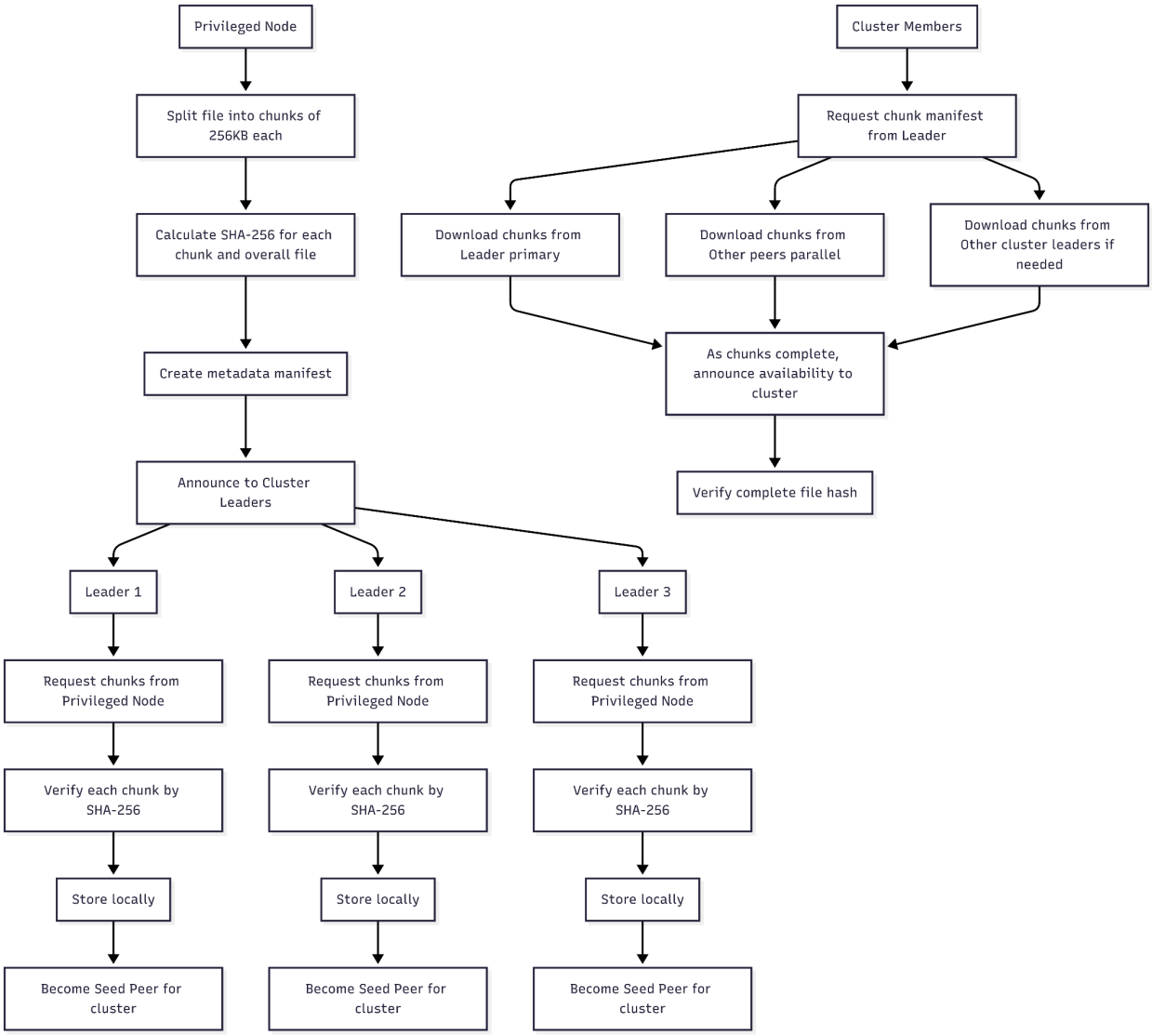
- After receiving a chunk:
  - Peer computes hash
  - Compares with original hash
- Corrupted chunks are re-requested

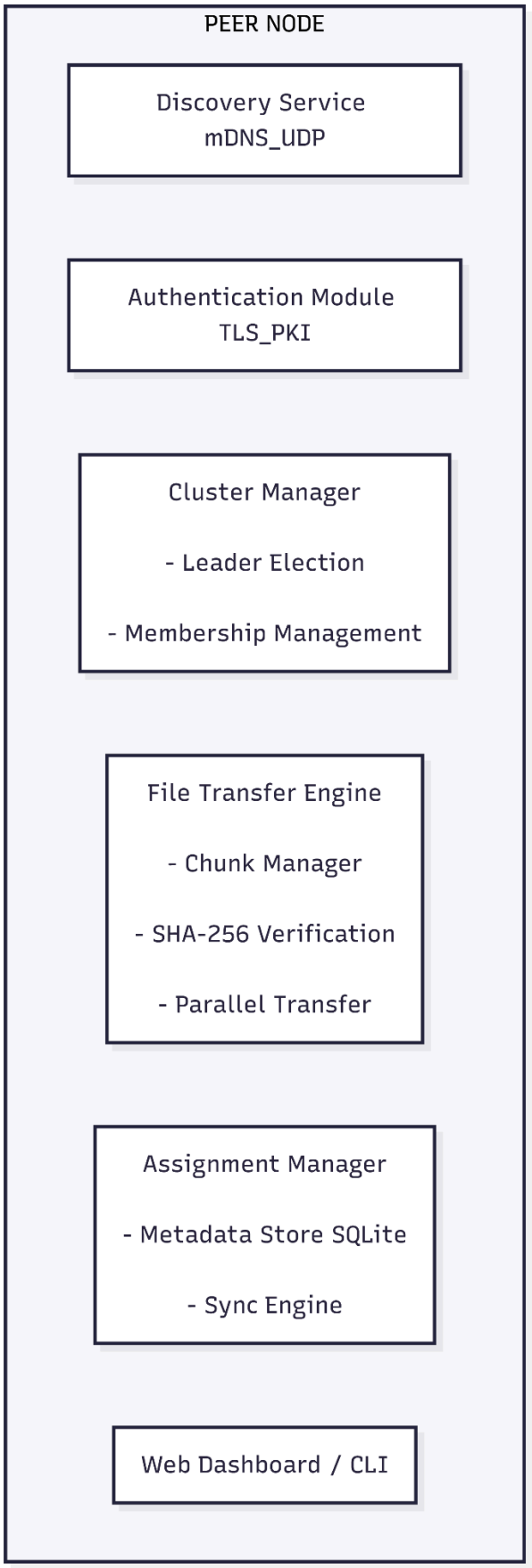
Step 7: Inter-Cluster Synchronization

- Cluster leaders exchange:
  - Missing chunks
  - Assignment metadata
- Ensures all clusters stay consistent

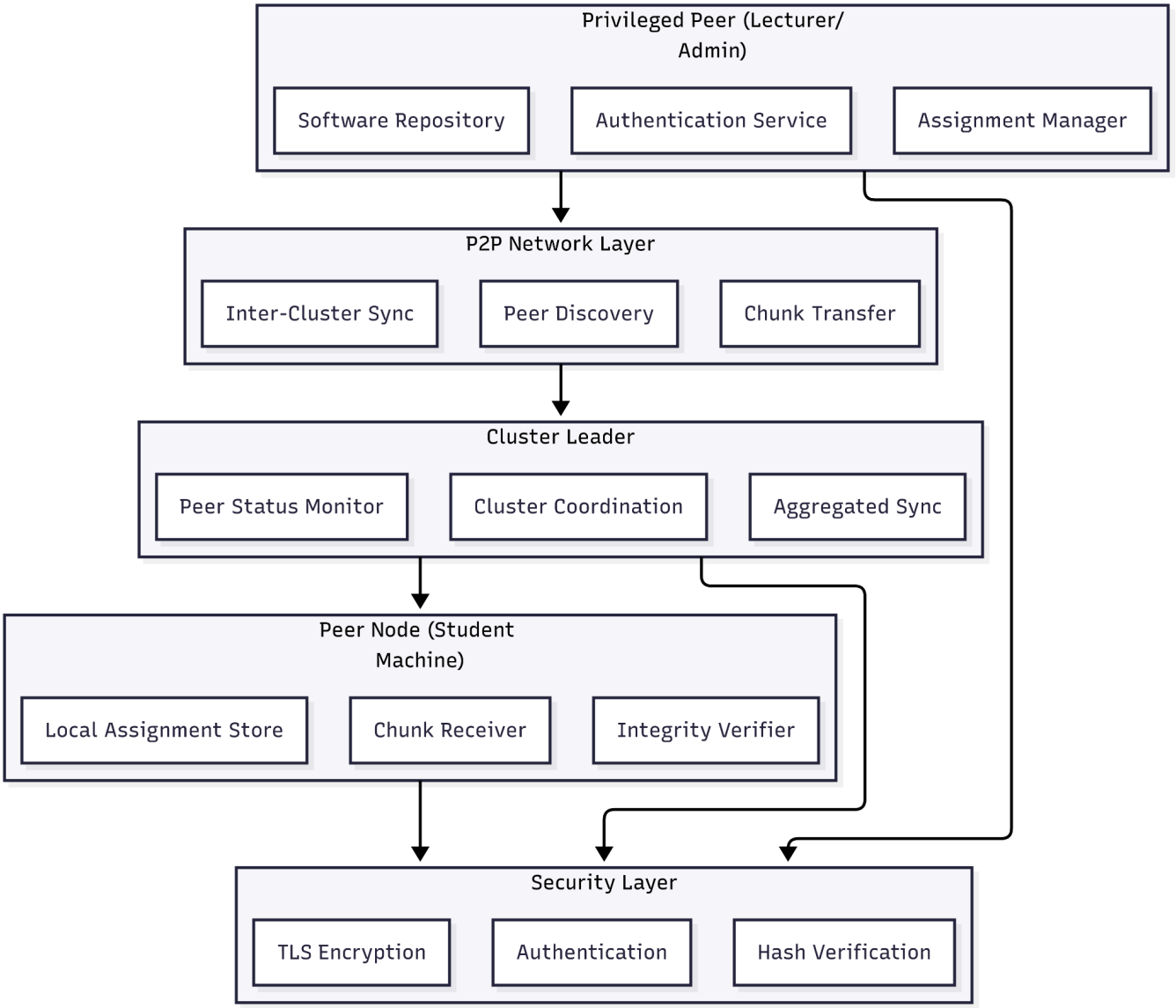
Step 8: Offline Support

- Offline peers:
  - Store assignments locally
  - Sync automatically when online
- Aggregated sync reduces network load

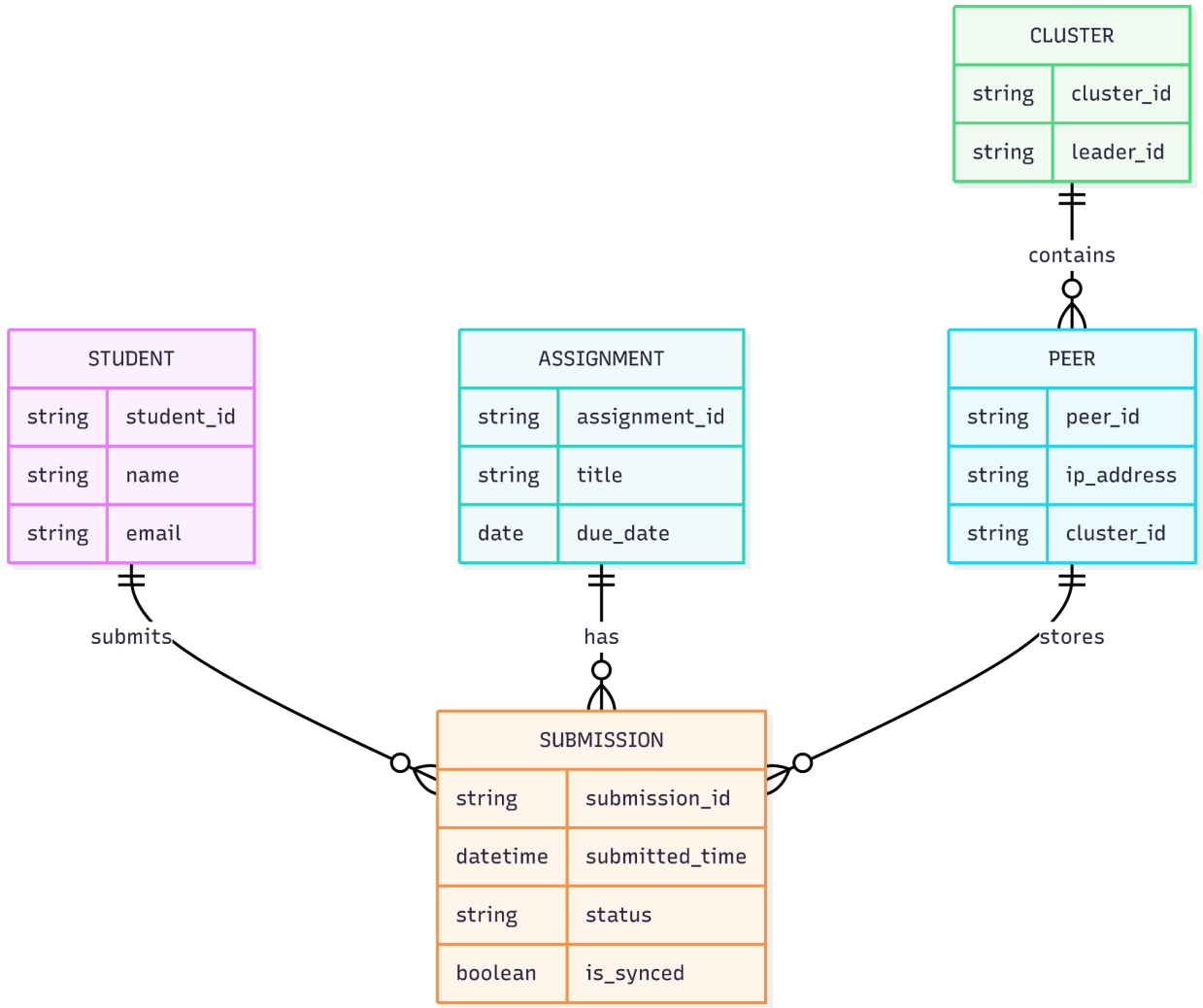

Privileged Node
Split file into chunks of 256KB each
Calculate SHA-256 for each chunk and overall file
Create metadata manifest
Announce to Cluster Leaders
Leader 1
Leader 2
Leader 3
Request chunks from Privileged Node
Verify each chunk by SHA-256
Store locally
Become Seed Peer for cluster
Cluster Members
Request chunk manifest from Leader
Download chunks from Leader primary
Download chunks from Other peers parallel
Download chunks from Other cluster leaders if needed
As chunks complete, announce availability to cluster
Verify complete file hash


PEER NODE
Discovery Service
mDNS_UDP
Authentication Module
TLS_PKI
Cluster Manager
- Leader Election
- Membership Management
File Transfer Engine
- Chunk Manager
- SHA-256 Verification
- Parallel Transfer
Assignment Manager
- Metadata Store SQLite
- Sync Engine
Web Dashboard / CLI

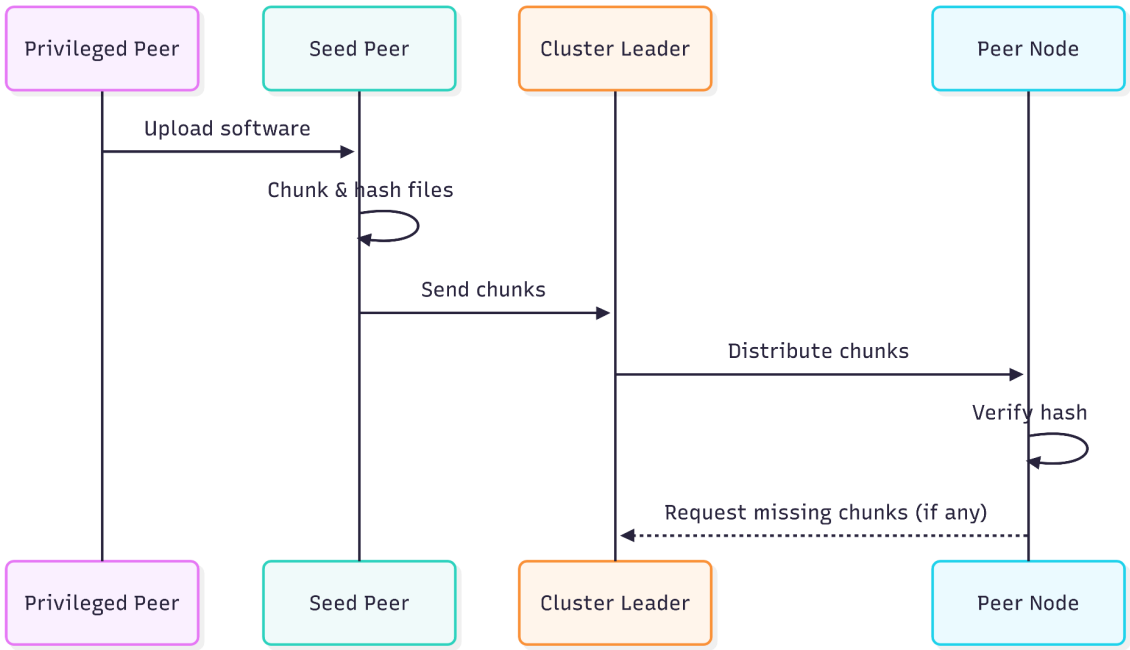
Overall Workflow

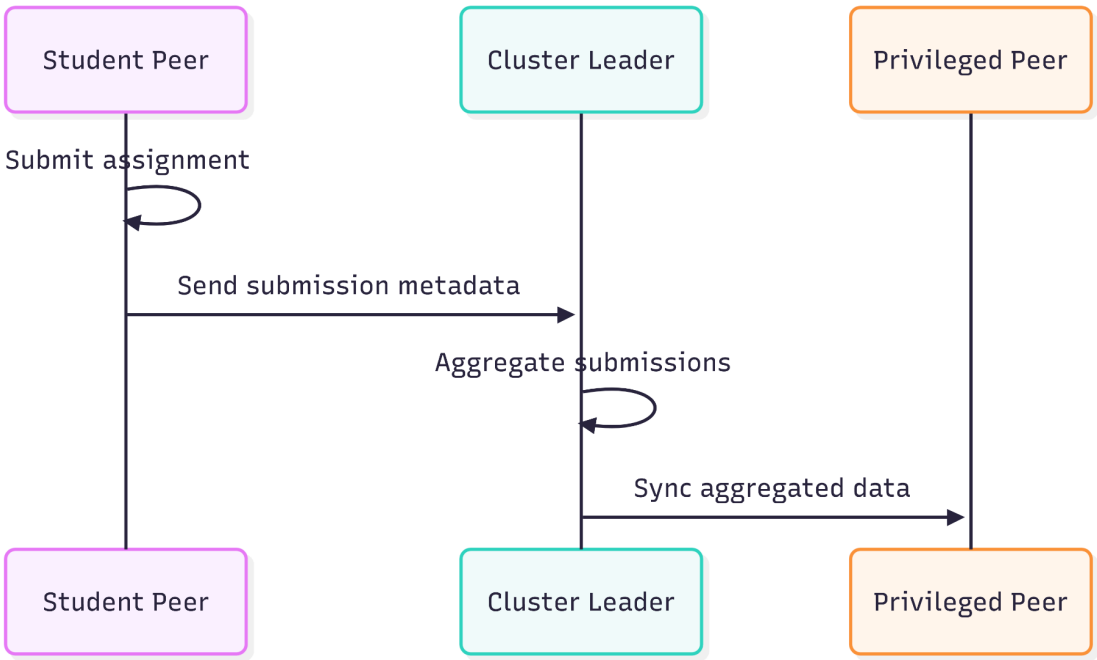


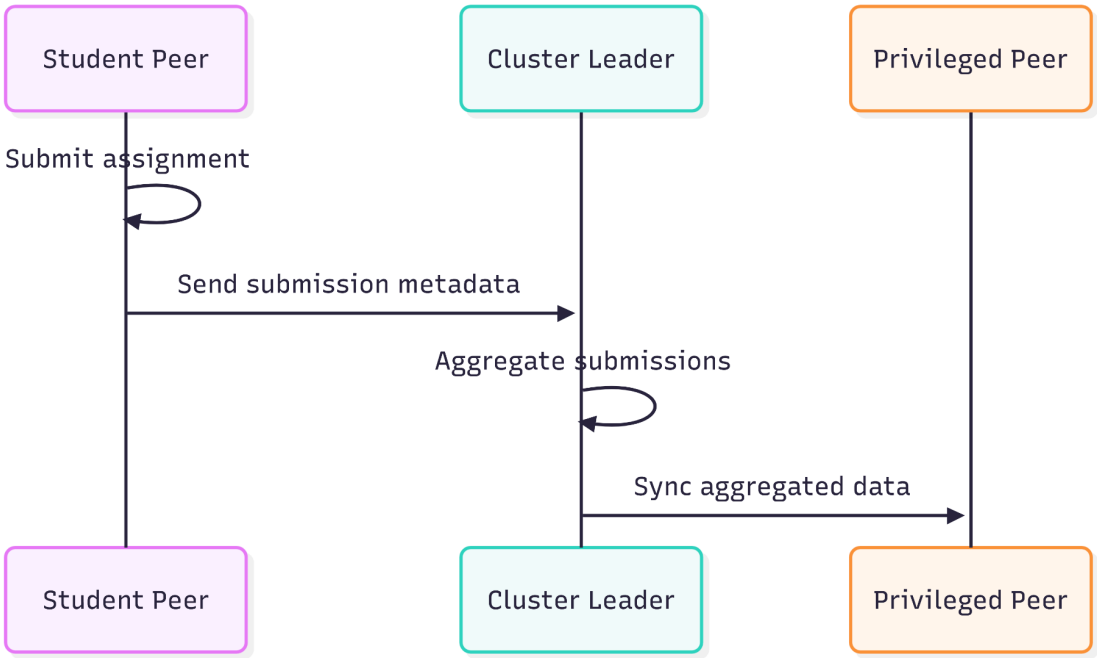
ER diagram for storage



Sequence Diagram






Student Peer
Cluster Leader
Privileged Peer
Submit assignment
Send submission metadata
Aggregate submissions
Sync aggregated data
Student Peer
Cluster Leader
Privileged Peer

Scenarios

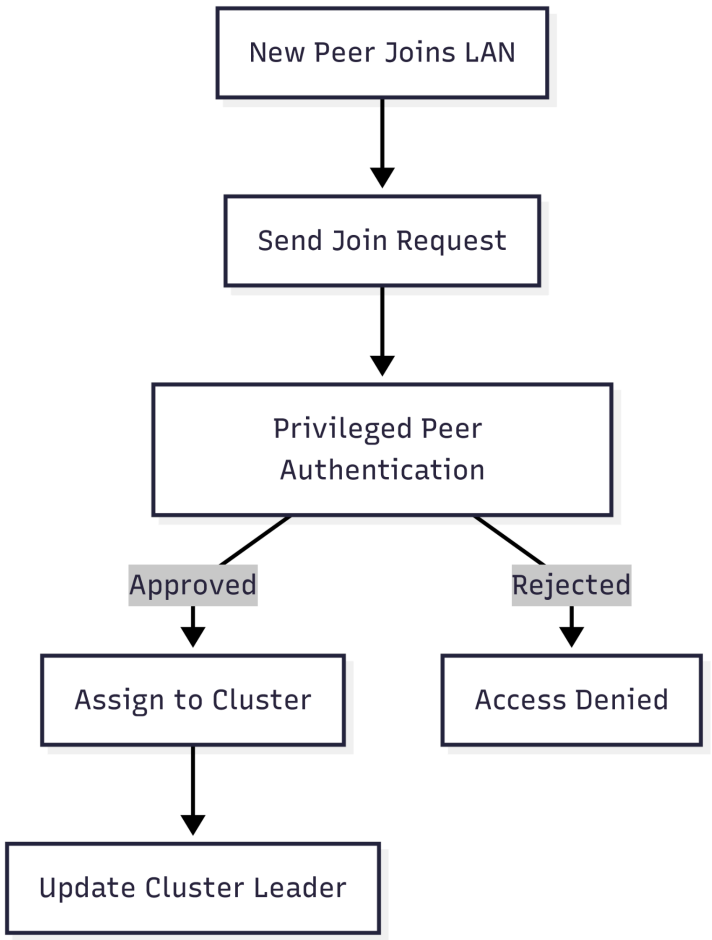
New Peer Joining the Network

Description

A student computer joins the lab network and becomes part of the P2P system.

Flow

1. Peer broadcasts join request
2. Privileged Peer authenticates
3. Peer assigned to a cluster
4. Cluster leader updates member list



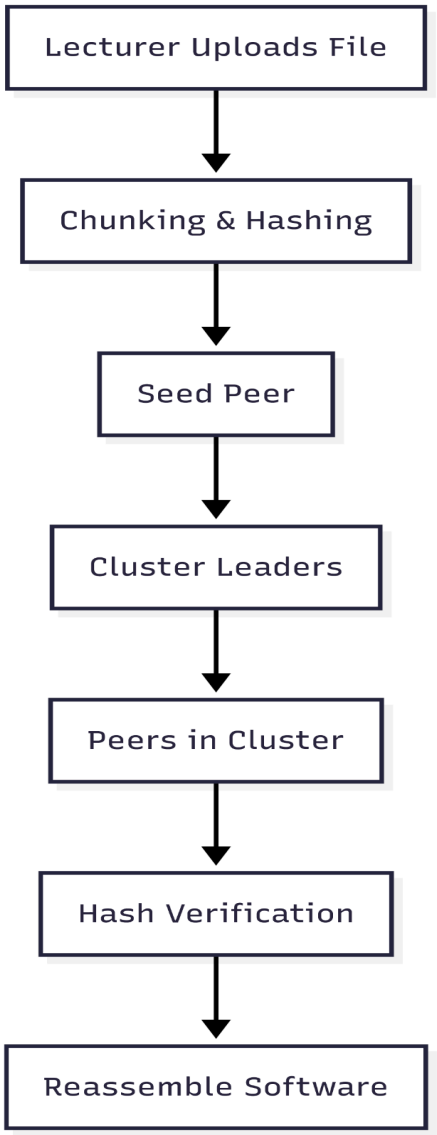
File/Software Distribution to Lab Machines

Description

Lecturer deploys software to all lab computers using P2P distribution.

Flow

1. Lecturer uploads software
2. Software split into chunks
3. Seed peer distributes chunks
4. Cluster leaders forward chunks
5. Peers verify and assemble software



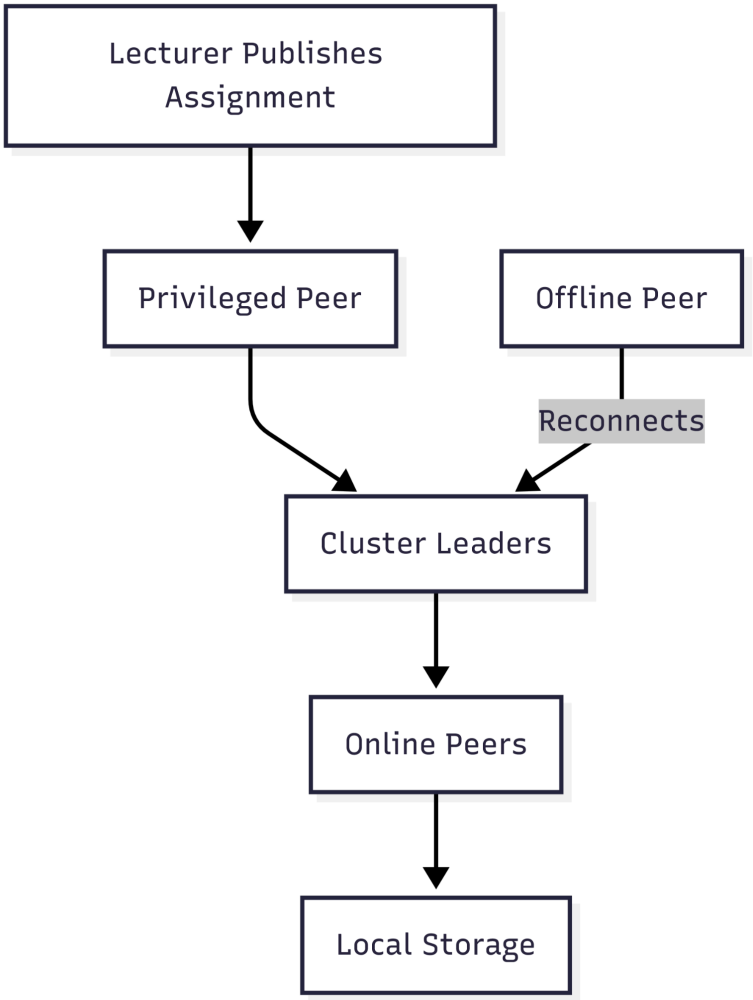
Assignment Distribution

Description

Assignments are distributed even if some students are offline.

Flow

1. Lecturer publishes assignment
2. Assignment metadata synced
3. Online peers receive data
4. Offline peers sync later



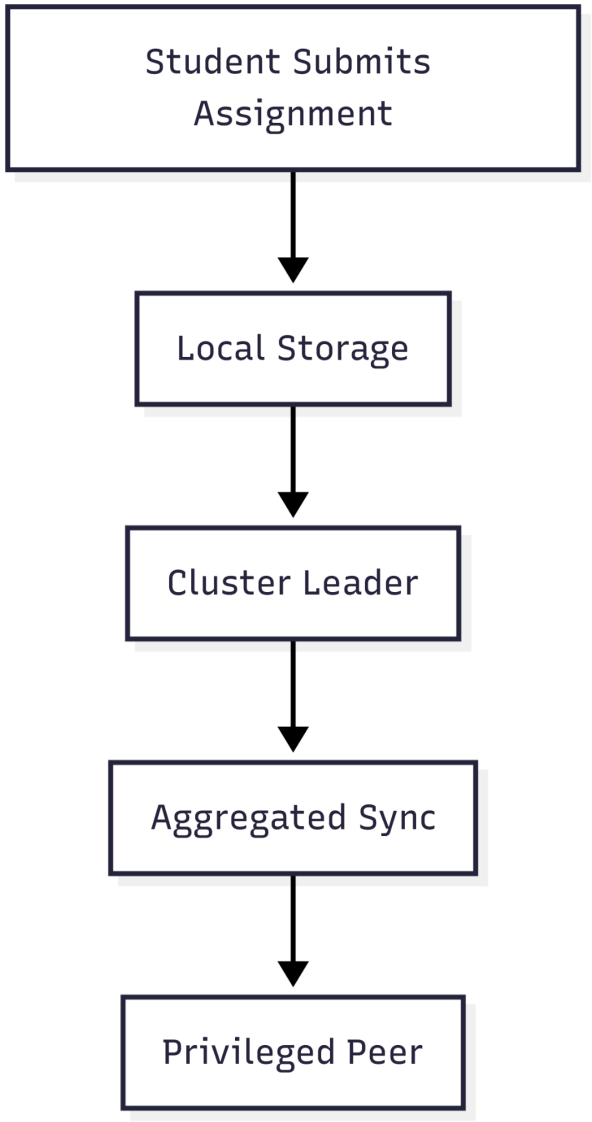
Assignment Submission & Aggregated Sync

Description

Students submit assignments; data is synchronized efficiently.

Flow

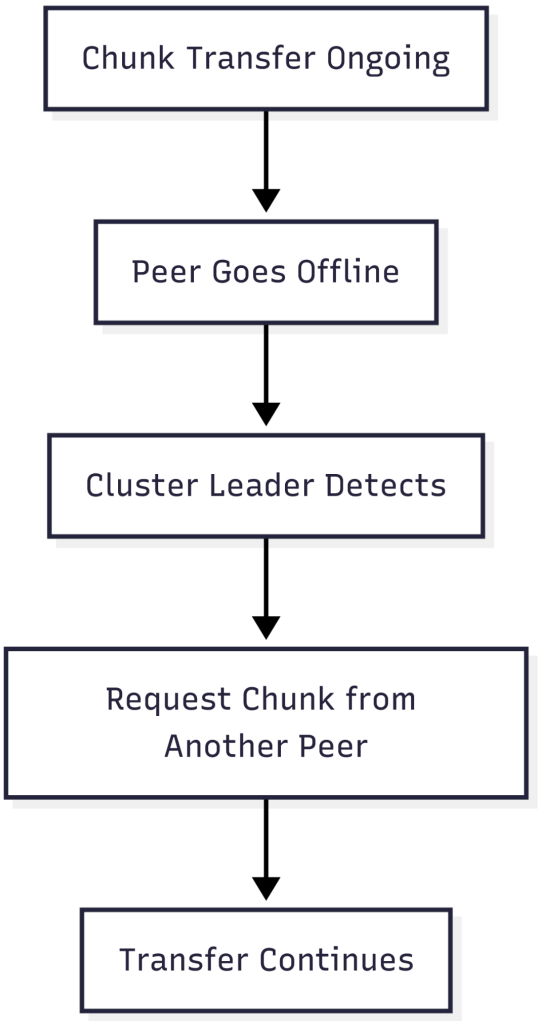
1. Student submits assignment
2. Stored locally if offline
3. Cluster leader aggregates submissions
4. Privileged Peer receives final data



Peer Goes Offline During Transfer

Description

A peer disconnects during file transfer.

Flow

1. Peer disconnects
2. Cluster leader detects failure
3. Another peer supplies missing chunks
4. Transfer continues